Hydrogen Diffusion in Fe-Mn-Al alloys

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SFB 761

7th Sept, 2017

CMD-AMS Scientific Retreat, Ebernburg, 4 - 7 Sept. 2017

Motivation

High-Mn austenitic steels

- Austenitic phase + high Mn content (18 – 28 wt.%) excellent combination of strength and ductility

- Addition of Aluminium

  - Reduces the specific weight Excellent candidate for manufacturing structural parts for automobiles

  - Increases the stacking fault energy Lowers the sensitivity to hydrogen induced delayed fracture[1]

- Presence of hydrogen

  - Degradation of mechanical properties (ductility) Hydrogen embrittlement

Chakrapani 1993

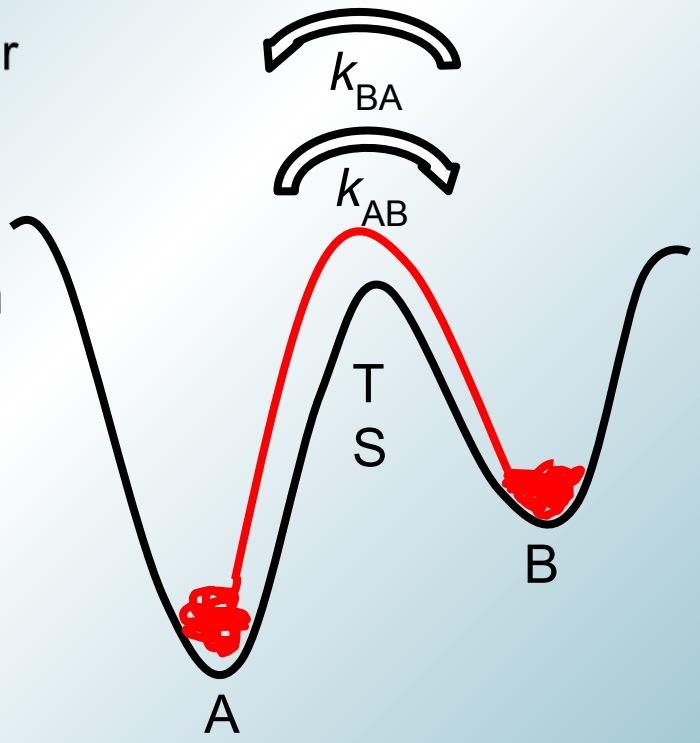
[1]Koyama *et al., International Journal of Hydrogen Energy, pg. 1-18, volume 42, 2017*

Key Aim

Effect of alloying element (Mn, Al) on H diffusivity in austenitic steels ?

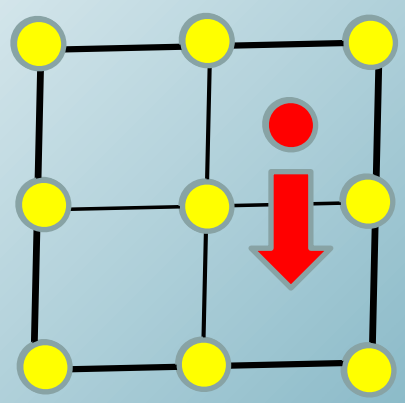
Method

- Diffusion takes place on a picosecond or longer timescale.
- "***Rare-events***" → time between consecutive events can be orders of magnitude longer than the event time itself.
- Molecular Dynamics → $\Delta t \sim 1$ fs



Kinetic Monte Carlo (kMC)

- Coarse grain the time evolution to the discrete rare events
- Trajectory consists of a sequence of discrete hops from site to site



kMC algorithm[2]

Start

Determine all possible processes *j* for a given system configuration

Get all rate constants k_j

Sum over all rate constants $k_{tot} = \sum_j k_j$

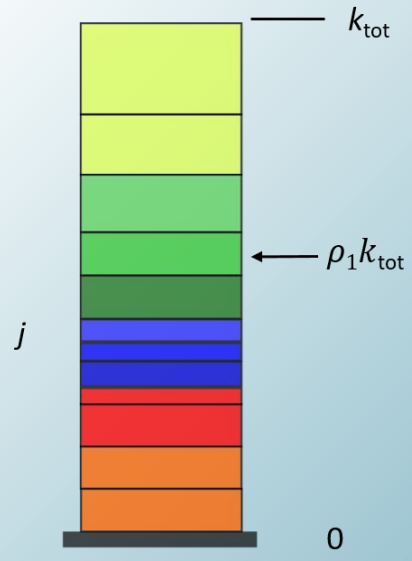
Get two random numbers $\rho_1, \rho_2 \in [0, 1]$

Select process "*i*" by $\sum_{j=0}^{i-1} k_j < \rho_1 k_{tot} \leq \sum_{j=0}^{i} k_j$

Execute process " *i* " and update configuration

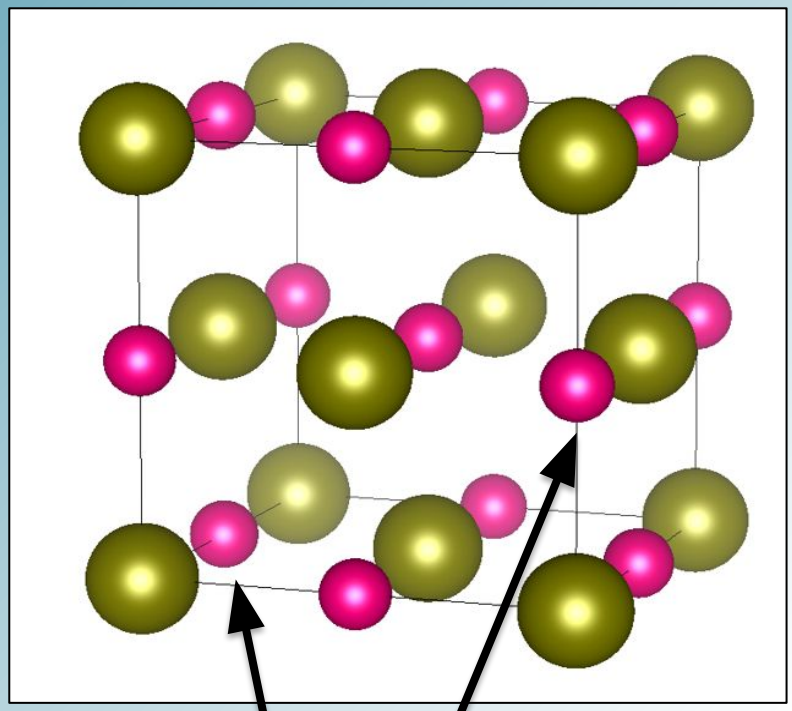
Advance time by $t \rightarrow t - \frac{\ln \rho_2}{k_{tot}}$

End

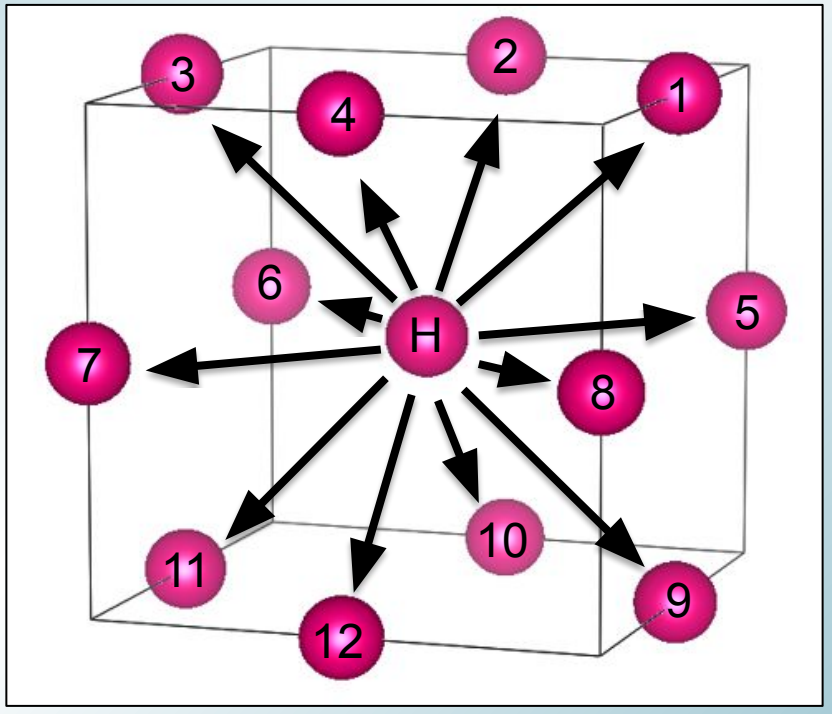


[2]Gillespie, J.Comp. Phys. 22, 403 (1976)

Model



Octahedral sites

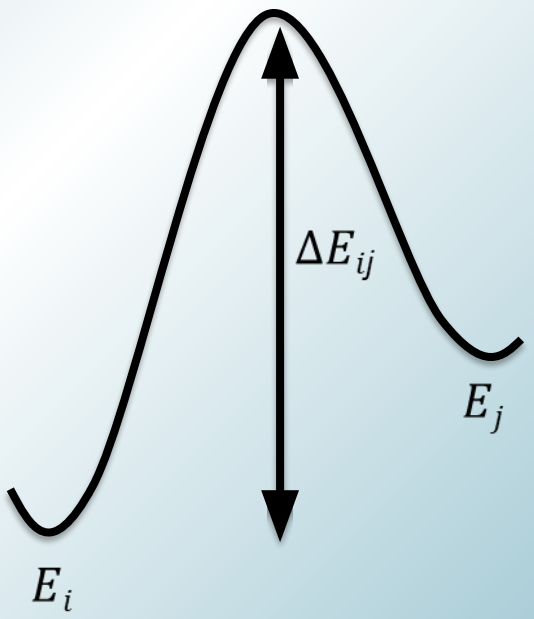


Possible hops (events)

- Each octahedral site surrounded by 12 nearest octahedral sites

Simulation Setup

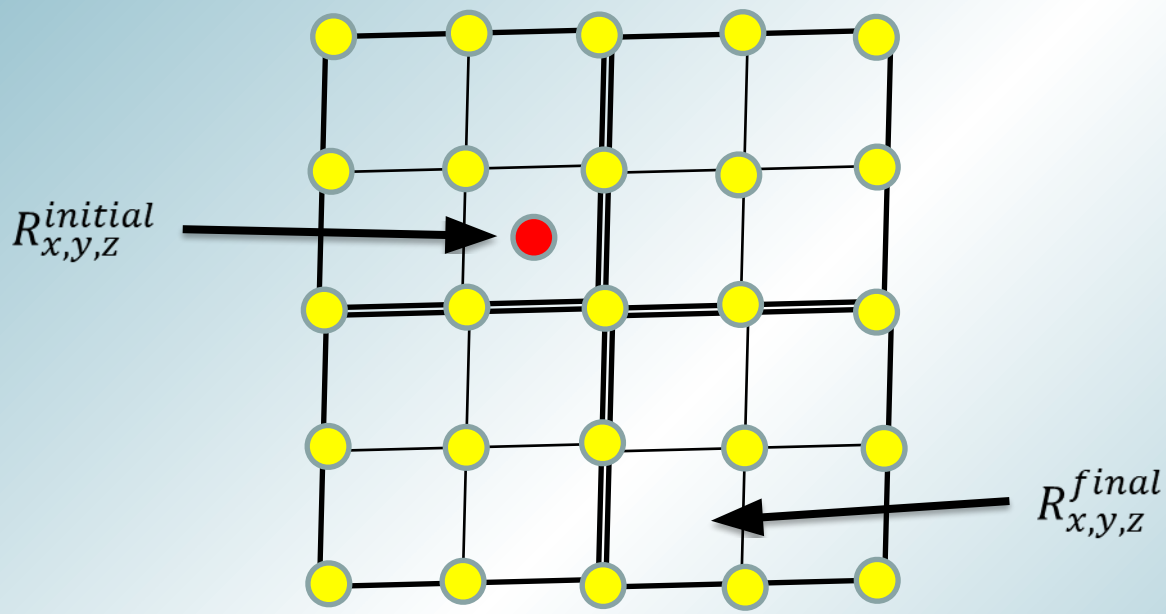
- System Size → 10×10×10 cubic unit cells
- Periodic boundary conditions in all three dimensions
- 4000 lattice sites and octahderal sites
- Convergence test
  - kMC steps – 10k per atom
  - 1000 H atoms
- Energy barriers extracted from DFT calculations[3]



- $E_i = E_i\,(N_{Mn}, N_{Al})$
- $E_j = E_j\,(N_{Mn}, N_{Al})$
- $\Delta E_{ij} = \Delta E_{ij}\,(N^i_{Mn}, N^i_{Al}, N^j_{Mn}, N^j_{Al})$

[3] Hüter *et al.,* Advances in Materials Science and Engineering, Vol. 2016, 9 pages, Calculations – E. McEniry

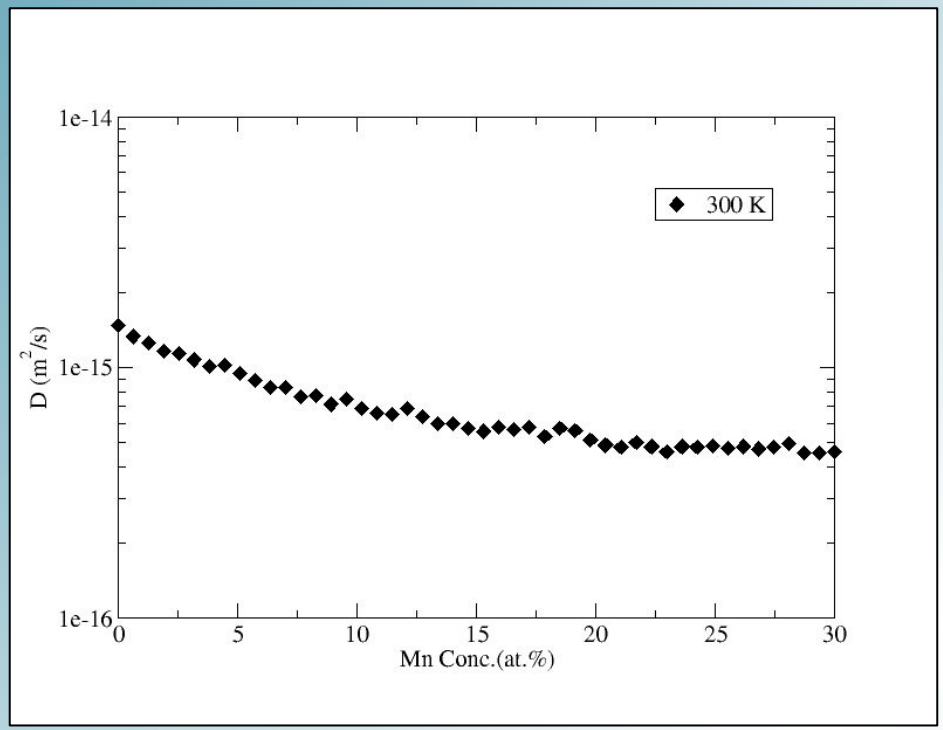
Diffusivity calculation

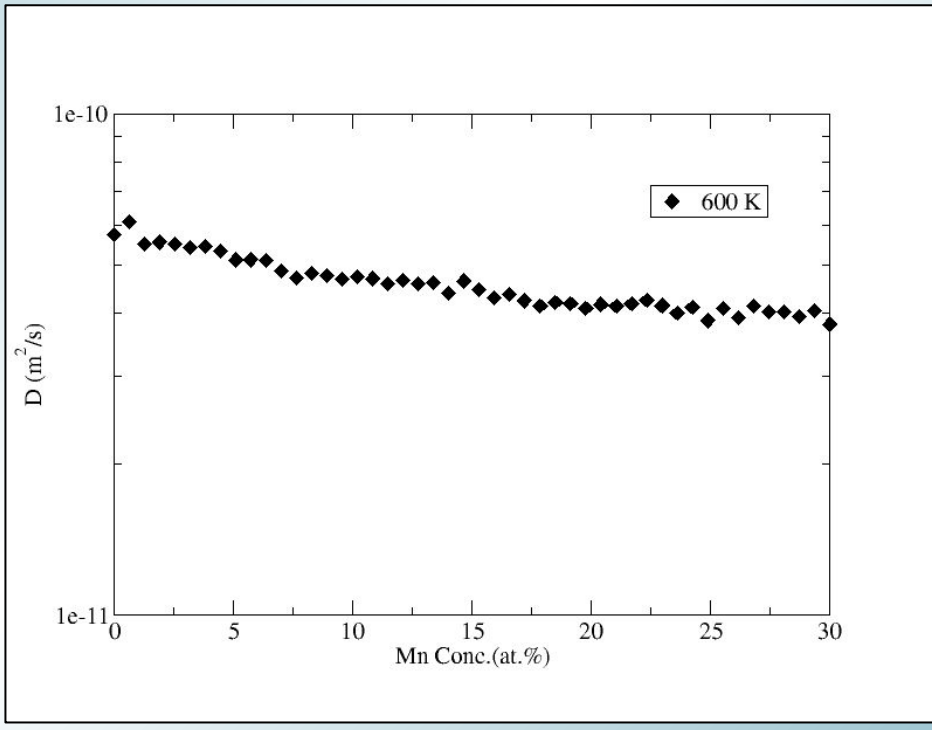


$$D_i = \frac{[R_{x,y,z}^{final} - R_{x,y,z}^{initial}]^2}{6\Delta t}$$

$$D = < D_i >$$

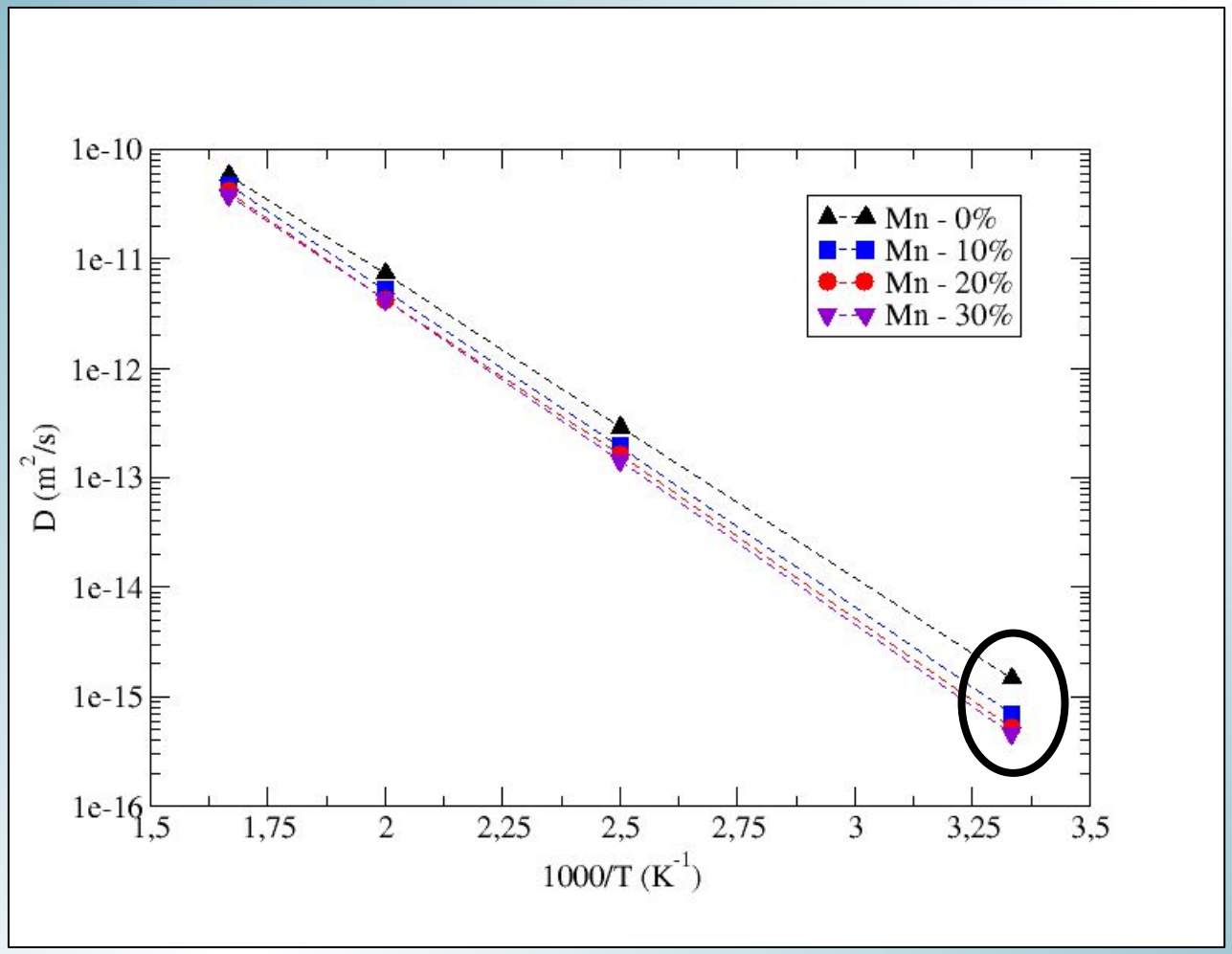
Effect of Mn content





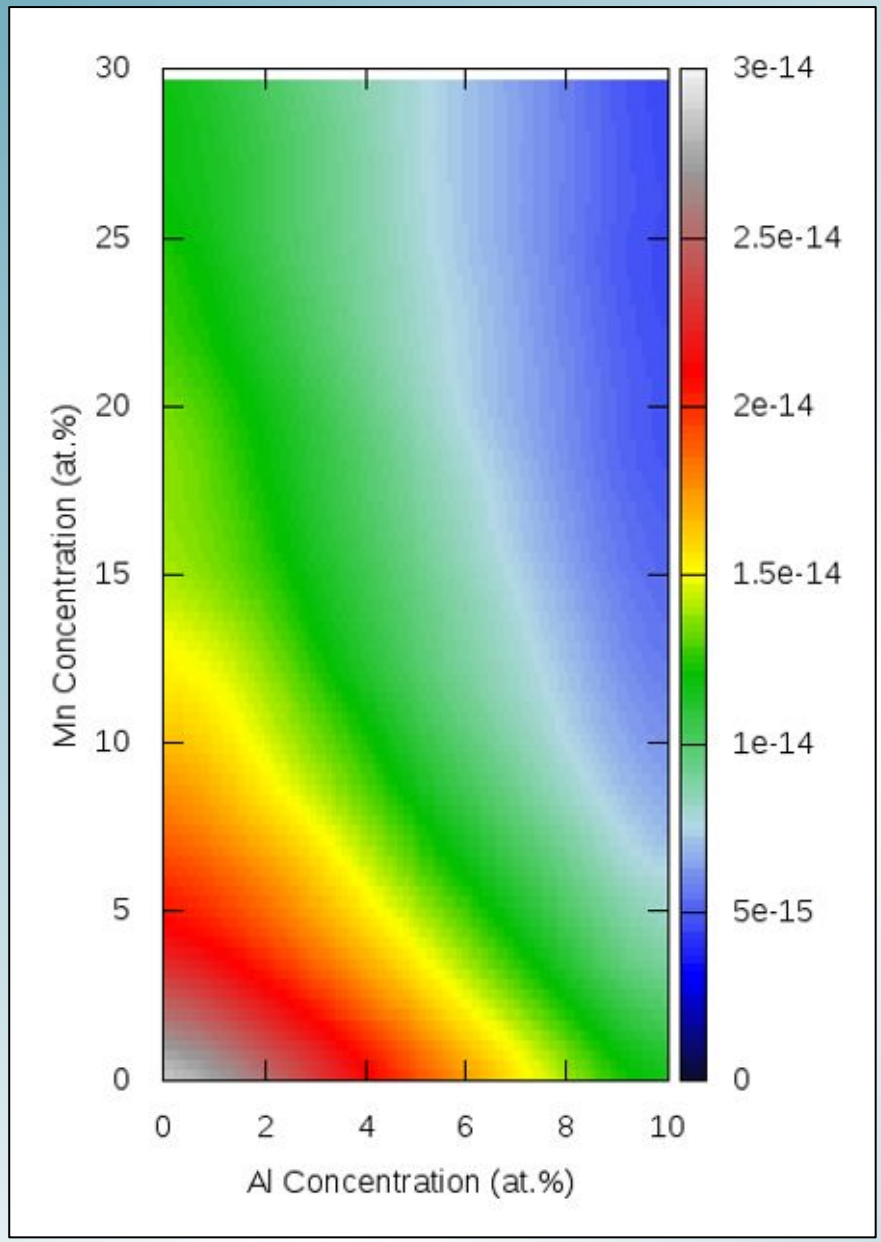
- H diffusivity decreases with increase in Mn content
- Change in H diffusivity is more at room temperature with the addition of Mn

Effect of Mn content



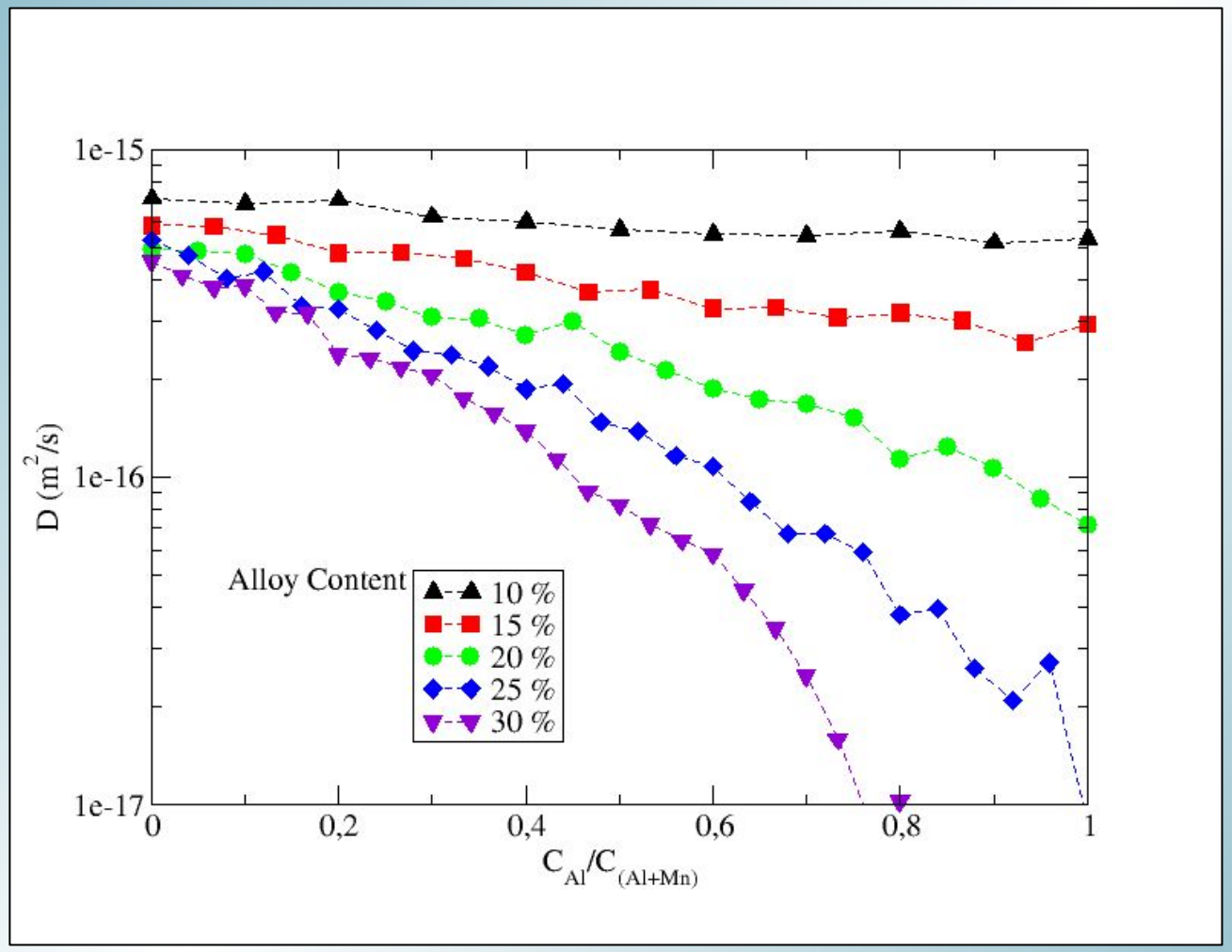
- Increase in Mn content from 0 at.% to 10 at.%. (room temperature) □ Significant drop in hydrogen diffusivity

Effect of Mn + Al content



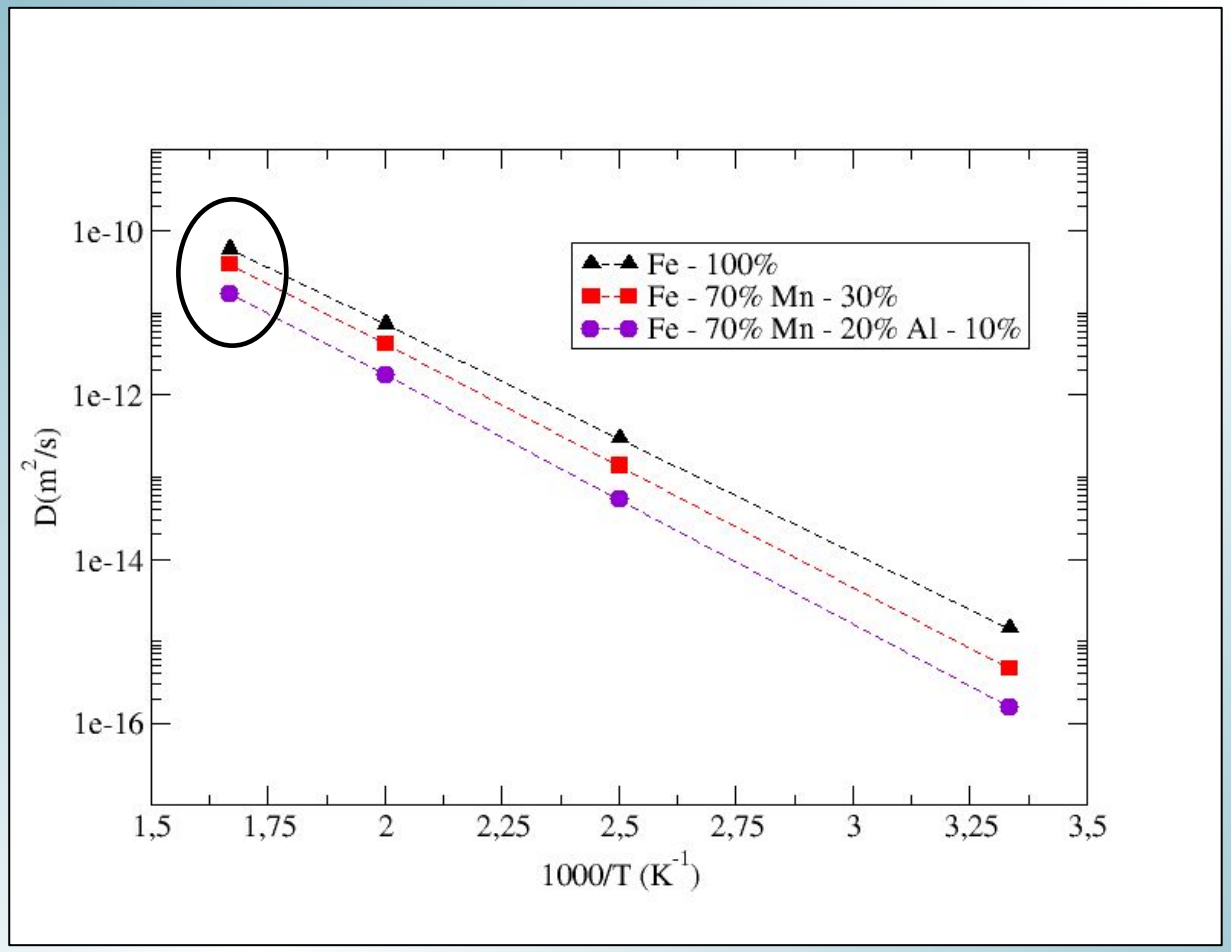
- Increase in Al content □ decrease in H diffusivity
- For the same decrease in H diffusivity, Al more effective than Mn

Effect of Mn + Al content



At higher concentrations of Al Model breaks down !

Combined Effect



Higher Temperature

- Addition of 10 at.% Al □ More effective

Summary and Conclusions

- A kinetic Monte Carlo code was developed to study the effect of alloying elements (Mn, Al) on H diffusion in austenitic steels.

Effect of alloying element (Mn, Al) on H diffusivity in austentic steels ?

- Alloying Mn and Al leads to decrease in hydrogen diffusivity.
- Decrease in H diffusivity is significant (factor of ~2) but not very large.
- Drop in diffusivity more evident at lower temperature Alloy effect more efficient.
- For a given alloy content, Al more efficient than Mn.

Outlook

- Segregation of Mn and Al at selected grain boundaries and how does it affects diffusivity.
- Correlated diffusion H-H interaction
- Inclusion of C in the interstitial sites (kappa - carbides) and it‘s effect on H diffusivity

Thank You For Your Attention